

TARANGINI 2.0

Diverse Dimensions of Knowledge

Editor
Dr Anandamma N



**Nagarjuna College of Management Studies
Chikkaballapur**

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Diverse Dimentions of Knowledge

**Editor
Dr. Anandamma N**

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Foreword

It gives me immense joy to introduce Tarangini 2.0, the second volume in our series of scholarly contributions in the field of STEM. This edition builds upon the strong foundation laid by its predecessor, carrying forward the spirit of inquiry, collaboration, and academic excellence that defines our institution.

Tarangini 2.0 is more than a collection of articles; it is a reflection of our continuous journey in advancing knowledge, embracing innovation, and encouraging interdisciplinary perspectives. The works featured here exemplify the intellectual curiosity and dedication of our faculty, researchers, and contributors, who remain committed to addressing contemporary challenges and exploring new horizons in science, technology, engineering, and mathematics.

This edition also signifies growth—a progression from ideas shared earlier to fresh insights that inspire deeper thought and wider impact. It stands as a testimony to our community's resilience, creativity, and determination to shape a better future through education and research.

I extend my sincere appreciation to all the contributors and the editorial team for their tireless efforts in bringing Tarangini 2.0 to life. May this volume continue to ignite curiosity, stimulate meaningful dialogue, and inspire generations of learners and leaders in STEM.

Warm regards,

Prof. Manohar Narajji
Director, Nagarjuna Education Society

Preface

It is with immense pride and great delight that I present Tarangini 2.0, the second edition of our collection of scholarly works in the STEM disciplines. Building on the success of the first volume, this edition reflects not only the intellectual depth of our faculty and collaborators but also our institution's continued commitment to fostering a culture of research, creativity, and academic excellence.

In an era defined by rapid advancements and ever-evolving challenges, the role of STEM education remains indispensable. The articles featured in Tarangini 2.0 embody this relevance—bringing together innovative research, practical perspectives, and forward-looking ideas that bridge theory with real-world application. Each contribution resonates with the spirit of inquiry and innovation that lies at the heart of progress.

What makes this volume particularly significant is the collaborative spirit it represents. Scholars from within our institution and distinguished experts from beyond have joined hands once again, demonstrating the power of knowledge-sharing and interdisciplinary exchange. Such collaborations enrich our academic fabric and open new avenues for impactful learning and discovery.

As you explore these pages, I hope you will be inspired by the diverse insights and thoughtful analyses presented. This volume is not just a publication but a reflection of our vision—to nurture minds that are curious, resilient, and prepared to lead in a world shaped by science, technology, engineering, and mathematics.

I extend my heartfelt gratitude to all the contributors and the editorial team for their dedication and tireless efforts in bringing Tarangini 2.0 to fruition. May this edition serve as

a source of inspiration, dialogue, and innovation for scholars and learners alike.

Warm regards,

Dr. Anandamma N
Principal, NCMS

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“AI as a Teaching Assistant: Transforming Education with Advanced Tools”

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Introduction

Artificial Intelligence (AI) is revolutionizing education by acting as a versatile teaching assistant. It makes administrative tasks easy, supports personalized learning, and enhances the teaching experience with cutting-edge tools. Teachers across various disciplines are leveraging AI-powered platforms to create engaging, effective, and accessible learning environments. This article explores AI tools designed for teaching and their applications in education, showcasing how technology helps both educators and students.

AI Tools for Teaching and Their Uses

1. Grammarly

Grammarly is a popular AI-powered tool that helps improve writing by providing real-time grammar, punctuation, and style suggestions. It is an effective tool for both students and educators, particularly when it comes to academic writing. Grammarly scans texts for errors and give suggestions to improve clarity, quality of the text and tone. For students, it acts as a virtual proof-reader, helping them refine essays, reports, and research papers. Teachers, on the other hand, can use it to assess the quality of student writing and provide immediate, constructive feedback. In addition to this, Grammarly helps students with writing skills by teaching them on what are the

common mistakes and improving their language usage over time.

2. Hemingway App

Hemingway App is another AI-driven writing tool that complements Grammarly by focusing on improving readability. It highlights complex sentences, excessive adverbs, and passive voice, making writing more effective and easier to understand. This tool is especially useful for students who may struggle with clear and direct writing. For teachers, Hemingway App can be used to simplify lesson plans, Course plan or study materials, by ensuring that complex ideas are communicated in an easy and effective manner.

3. Duolingo

Duolingo is a effective AI tool that has revolutionized the way students learn new languages. It uses gamification techniques to make learning a fun and interactive experience. Duolingo adapts to the learner's proficiency, offering personalized lessons that helps student's improve skills. It also helps students in learning languages outside the classroom, as it provides immediate feedback, reinforces learning through repetition, and tracks progress. Educators can incorporate Duolingo into their language programs where they allow students to practice outside of formal lessons, helping them build fluency and confidence in the target language.

4. Labster

For practical and hands-on learning, especially in STEM subjects, Labster provides virtual lab simulations that allow students to conduct experiments in subjects like biology, chemistry, and physics. Labster brings the laboratory experience to life through interactive, immersive simulations that replicate real-world experiments without the need for physical lab equipment.

This is especially useful in educational settings with limited

resources or where access to a physical laboratory is not feasible. Students can explore scientific concepts safely and at their own pace, while teachers can monitor student progress and provide guidance based on their virtual lab work. Labster enhances students' understanding of complex scientific theories by offering them the freedom to experiment and learn from their mistakes in a risk-free environment.

5. Kahoot!

Kahoot! is another AI-powered tool that engages students through gamified learning experiences. It enables educators to create quizzes, surveys, and polls that can be played in real-time, either individually or in teams. By turning assessments into competitive games, Kahoot! increases student participation and enthusiasm, making learning more enjoyable. This tool is especially effective for reviewing concepts, reinforcing knowledge, and conducting formative assessments. Educators can use Kahoot! to gauge understanding quickly, provide instant feedback, and ensure that students remain actively involved in the learning process.

6. Quizlet

Quizlet is an AI-driven tool that enables students to create flashcards and study sets for various subjects. It also offers pre-made study sets created by other users. Through spaced repetition and quizzes, students can remember the key topics. Teachers can use Quizlet to create customized flashcards, review activities, and track student progress. It is particularly useful for subjects that require memorization, such as languages, history, and science.

7. Google Classroom

Google Classroom is a learning management system (LMS) that integrates with Google's set of productivity tools, helping educators in manage assignments, grades, and communication. With AI, Google Classroom helps teachers track student

progress, provide instant feedback, and streamline administrative tasks. Students can submit assignments, participate in discussions, and access resources all in one place. The tool makes collaboration easier and more efficient, allowing for real-time updates and seamless communication between teachers, students, and parents.

8. Tynker

Tynker is a coding platform designed for children and beginners to learn programming. Using interactive lessons and AI, Tynker introduces coding concepts like loops, conditions, and algorithms in a fun and engaging way. It allows students to create their own games, animations, and apps. Teachers can use Tynker to introduce coding in a classroom setting, guiding students through step-by-step tutorials and offering hands-on experience with programming in a creative environment.

9. Turnitin

Turnitin is a widely used AI tool for detecting plagiarism in student assignments. It compares submitted work against a vast database of academic articles, books, and other student papers to identify potential instances of copied content. In addition to plagiarism detection, Turnitin offers feedback on writing quality and citation practices. Educators use it to maintain academic integrity and help students learn the importance of proper referencing and original work.

10. Microsoft Immersive Reader

Microsoft Immersive Reader is an AI tool designed to assist students with reading comprehension and accessibility. It offers features like text-to-speech, translation, and adjustable text spacing, making it particularly helpful for students with learning disabilities or language barriers. Teachers can use the tool to support students who struggle with reading or need accommodations, ensuring all learners have equal access to educational content.

11. Canva for Education

Canva for Education is a design tool that allows students and teachers to create visually appealing presentations, infographics, posters, and other learning materials. With AI- powered templates, students can design professional-quality visuals that help communicate their ideas effectively. Educators can use this Canva to create education related content and students can use it to showcase their understanding in creative projects, fostering visual literacy and creativity.

12. Mentimeter

Mentimeter is an interactive tool that allows educators to gather real-time feedback from students through polls, quizzes, and open-ended questions. Using AI, Mentimeter generates visual reports based on the data collected, helping teachers gauge understanding and engagement. It is an excellent tool for making lessons more interactive, encouraging participation, and adapting teaching strategies based on student input. Students can respond anonymously, creating an open space for honest feedback.

Conclusion

AI tools are reshaping education by acting as indispensable teaching assistants, offering innovative solutions for lesson delivery, student engagement, and academic administration. From writing assistance to virtual labs and gamified learning platforms, these tools enhance the educational experience across all disciplines. While challenges like ensuring equal access and ethical use remain, the potential of AI to revolutionize teaching is undeniable. With thoughtful integration and responsible use, AI can empower educators and create a more inclusive, dynamic, and effective learning environment for all.

THE FUTURE OF ARTIFICIAL INTELLIGENCE

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Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and problem-solve. In recent years, AI has progressed rapidly, evolving from simple automated systems to complex algorithms capable of performing tasks that were once exclusive to humans, such as driving cars, diagnosing diseases, and managing financial portfolios. AI is now an integral part of various industries, from healthcare and education to finance and entertainment. As we look to the future, the question arises: How will AI shape society? While AI promises tremendous benefits, it also presents significant ethical dilemmas, challenges to the workforce, and shifts in societal structures.

AI brings with it a host of ethical concerns that need to be addressed as it continues to develop. One of the most pressing issues is bias in algorithms. AI systems learn from data, and if the data used to train these systems is biased, the AI can inadvertently perpetuate those biases. For example, if an AI algorithm used in hiring is trained on data that reflects past discriminatory hiring practices, the system may unfairly favor certain candidates over others. This can exacerbate existing inequalities in society, particularly regarding race, gender, and socioeconomic status.

Another significant ethical dilemma is privacy. AI systems often rely on vast amounts of personal data to function effectively. Whether it's tracking online behaviour, analyzing medical

histories, or monitoring people's daily activities through smart devices, AI systems are capable of collecting and processing intimate details about individuals' lives. The question arises: How much of this data is too much? And who is responsible for protecting it? With data breaches becoming increasingly common, the need for robust privacy protections is more critical than ever.

Accountability is another crucial ethical consideration. When an AI system makes a decision—be it in healthcare, law enforcement, or finance—who is held responsible if that decision leads to harm? For instance, if an autonomous vehicle causes an accident, is the manufacturer liable? Or is it the software developer who created the driving algorithm? Establishing clear accountability for AI decisions is essential to ensuring fairness and preventing harm.

As AI continues to advance, one of the most significant concerns is its impact on the workforce. On one hand, AI automation can create new jobs in emerging tech fields. As AI technology develops, there will be a growing demand for data scientists, AI specialists, software engineers, and cybersecurity experts to design, maintain, and safeguard these systems. Additionally, AI could lead to the creation of entirely new industries, much like how the internet gave rise to e-commerce, social media, and digital marketing. These new fields will require highly skilled workers, and thus, AI could become a catalyst for job creation in the tech sector.

However, the negative impact on jobs cannot be overlooked. Automation powered by AI is expected to replace many roles, particularly in industries such as manufacturing, transportation, and customer service. Self-driving vehicles, for instance, could displace millions of truck drivers, while AI-powered chatbots could replace customer service representatives.

In healthcare, AI has the potential to revolutionize patient care. Machine learning algorithms can analyze vast amounts of medical data, leading to more accurate diagnoses, personalized treatment plans, and improved patient outcomes. AI systems can also assist in drug discovery, accelerating the development of life-saving treatments. However, AI's role in healthcare must be approached with caution, as it could raise concerns about data privacy, patient consent, and the dehumanization of care.

In education, AI can personalize learning by adapting lessons to individual students' needs, improving the efficiency of teaching, and helping educators better understand how students learn. AI-powered tools can also assist in administrative tasks, freeing up time for teachers to focus on instruction. On the flip side, there are concerns about the digital divide, where students in low-income areas may not have access to the same AI-powered resources, thus exacerbating inequalities in education.

Personal relationships could also change with AI's integration into daily life. Virtual assistants, chatbots, and AI-driven social media platforms already mediate our interactions with others. While these technologies can help us stay connected, they can also lead to a decline in meaningful face-to-face interactions. The rise of AI-powered companions, such as robots or virtual friends, could further complicate human relationships, blurring the lines between human connection and machine interaction. As AI continues to evolve, society faces the challenge of balancing its immense potential with responsible governance. Ethical concerns, such as bias, privacy, and accountability, must be addressed to ensure that AI systems are fair, transparent, and secure. The workforce will undergo significant transformation, with new opportunities in tech fields alongside job displacement in traditional industries. The key to mitigating these changes will be retraining workers and fostering a flexible labor market that embraces the future of work. Finally, the impact of AI on society will depend on how it is integrated into our lives—

whether it is used to enhance human connection, improve healthcare, and revolutionize education or whether it exacerbates inequalities and disrupts personal relationships.

To fully harness the benefits of AI while mitigating its risks, society must establish clear regulations, invest in education and training, and ensure that AI development prioritizes human well-being. By doing so, we can ensure that AI remains a tool for good, driving progress without sacrificing the values that make us human.

Diamonds: The Future of Ultra-Stable Data Storage

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From the earliest stone tools to the latest advances in artificial intelligence, robotics, blockchains and virtual reality, technology has continuously transformed our lives. Stay informed about the latest technological breakthroughs and the ethical and societal norms they challenge.

Record-breaking diamond storage can save data for millions of years. Researchers have used lasers to encode information in diamonds, demonstrating record-breaking data density in an ultra-stable and long-lasting system.

The famous marketing slogan about how a diamond is forever may only be a slight exaggeration for a diamond-based system capable of storing information for millions of years – and now researchers have created one with a record-breaking storage density of 1.85 terabytes per cubic centimetre.

Previous techniques have also used laser pulses to encode data into diamonds, but the higher storage density afforded by the new method means a diamond optical disc with the same volume as a standard Blu-ray could store approximately 100 terabytes of data – the equivalent of about 2000 Blu-rays – while lasting far longer than a typical Blu-ray's lifetime of just a few decades.

“Once the internal data storage structures are stabilised using our technology, diamond can achieve extraordinary longevity – data retention for millions of years at room temperature – without requiring any maintenance,”

Wang and his colleagues worked with small pieces of diamond only a few millimetres long, although they say future versions of the system could be in the form of larger storage discs. Their method used ultrafast laser pulses to knock some of a diamond's carbon atoms out of place, leaving behind empty spaces the size of single atoms that each exhibited a stable brightness level.

By controlling the energy of the laser, the researchers could make multiple empty spaces at specific sites within the diamond, and the density of those spaces influenced each site's overall brightness. "The number of empty spaces can be determined by looking at the brightness, which allows us to read the stored information," says Wang.

The team then stored images – including Eadweard Muybridge's 1878 sequence of photos showing a rider on a galloping horse – by mapping the brightness of each pixel to the brightness levels of specific sites inside the diamond. The system saved this data with more than 99 per cent accuracy and completeness.

This storage method isn't yet commercially viable because it requires expensive lasers and high-speed fluorescence imaging cameras, along with other devices, says Wang. But he and his colleagues expect that their diamond-based system could eventually be miniaturised to fit within a space the size of a microwave oven.

"In the short term, government agencies, research institutes and libraries focused on archiving and data preservation would likely be eager to adopt this technology,"

APPLICANT OF TRACKING SYSTEM

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INTRODUCTION :-

An Applicant Tracking System (ATS) is a software tool designed to simplify and streamline the recruitment process for organizations. It helps employers manage job postings, collect and organize resumes, screen candidates, and track their progress throughout the hiring pipeline. Whether used by small businesses or large corporations, an ATS has become an essential tool in modern recruitment.

Applicant Tracking System :-

An Applicant Tracking System is a digital platform designed to parse, store, and analyze applicant data, enabling recruiters to filter and rank candidates based on predefined criteria such as skills, experience, and qualifications. Its primary purpose is to streamline and automate the recruitment process.

Key Features of ATS :-

1. Resume Parsing

Automatically extracts key details (skills, experience, education) from resumes, making it easier to filter and shortlist candidates without manual effort.

2. Job Posting & Distribution

Allows recruiters to post job openings on multiple platforms (company website, job boards, social media) from a single system, increasing visibility and reach.

3. Automated Candidate Screening

Uses AI or keyword-based filters to rank applicants based on job requirements, helping recruiters quickly identify the most qualified candidates.

4. Interview Scheduling

Integrates with calendar systems (Google, Outlook) to schedule interviews automatically, reducing back-and-forth communication between recruiters and candidates.

5. Collaboration & Feedback Management

Enables hiring teams to leave comments, rate candidates, and share feedback in a centralized system, improving team decision-making.

6. Reporting & Analytics:

Generates insights on recruitment performance (time-to-hire, source effectiveness, diversity metrics) to help optimize hiring strategies.

How It works :-

1. Job Posting:

The company creates a job posting and publishes it on their website, job boards, or social media.

2. Applicant Submission:

Candidates submit their resumes and cover letters through the company's website, email, or job boards.

3. ATS Screening:

The ATS software screens and filters resumes based on keywords, qualifications, and other criteria set by the hiring manager.

4. Ranking and Sorting:

The ATS ranks and sorts candidates based on their relevance to the job posting.

5. Candidate Profile Creation:

The ATS creates a candidate profile, which includes their resume, cover letter, and other relevant information.

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6. Hiring Manager Review:

The hiring manager reviews the top-ranked candidates and selects those to move forward in the hiring process.

7. Interview Scheduling:

The ATS can also be used to schedule interviews and send reminders to candidates.

Benefits of Using an ATS :-

Time Efficiency

Automating repetitive tasks like resume screening and interview scheduling.

Improved Candidate Experience

Faster communication and updates for applicants.

Better Candidate Matching

Identifying the most qualified candidates based on job requirements.

Data-Driven Decisions

Using analytics to improve recruitment strategies.

Compliance

Ensuring adherence to hiring laws and regulations.

Types of Applicant Tracking System :-

1. Enterprise ATS

Designed for large organizations with high-volume hiring.

Features include AI-driven candidate screening, integrations with HR software, and multi-location support.

Example: Workday, SAP SuccessFactors

2. Small Business ATS

Tailored for startups and small businesses with simpler hiring

needs. Offers basic resume parsing, job posting, and candidate tracking.

Example: Breezy HR, Workable

3. Cloud-Based ATS (SaaS ATS)

Hosted online and accessible from anywhere. Usually subscription-based and scalable.

Example: Greenhouse, Lever

4. On-Premise ATS

Installed on company servers for maximum data control.

Preferred by organizations with strict data security policies.

Example: IBM Kenexa

5. Industry-Specific ATS

Designed for specific sectors like healthcare, IT, or retail. Includes customized workflows for industry-specific hiring processes.

Example: Bullhorn (for staffing agencies)

6. Open-Source ATS

Allows companies to modify and customize the software as needed. Requires technical expertise for deployment and maintenance.

Example: OpenCATS, iKrut

7. AI-Powered ATS

Uses artificial intelligence to automate resume screening, chatbots for candidate interaction, and predictive analytics.

Example: HireVue, Pymetrics.

Future Trends in Applicant Tracking Systems :-

- AI and Machine Learning: Enhanced candidate matching and predictive analytics.
- Mobile Optimization:- Improving mobile accessibility for recruiters and candidates.

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- Diversity and Inclusion Tools: Reducing bias and promoting equitable hiring practices.
- Integration with Virtual Hiring Tools: Supporting remote interviews and onboarding.

CONCLUSION:-

An Applicant Tracking System (ATS) streamlines the recruitment process by automating job postings, resume screening, and candidate management. It enhances efficiency, reduces hiring time, and improves the overall quality of hires by ensuring that only the most qualified candidates advance in the selection process. By leveraging data-driven insights and optimizing workflows, an ATS helps organizations make informed hiring decisions while enhancing the candidate experience. Adopting an ATS is essential for modern businesses seeking to attract, assess, and hire top talent effectively.

CYBER SECURITY

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In order to securely guarantee the safety and security of information systems, cyber security has grown to be a highly important issue that requires the attention of researchers, academicians, and corporations. Every person and business must contend with constantly evolving cyberthreats as a result of the growing demand for digitalization. An overview of the present state of cyber security, its difficulties and strategies, as well as its global trends, is given in this article. We carried out a comprehensive review to find the most recent trends, difficulties, and state-of-the-art in cyber security in order to stay ahead of the curve. Additionally, we discuss the future of cyber security, including potential tactics and methods to deal with the growing threat landscapes, new developments, and advancements such as artificial intelligence (AI). In fact, during the past few decades, the internet has grown rapidly, radically altering vital infrastructure, the economy, and society. Often called the "cyber civilization," this digital revolution has completely changed how we access information, interact, and do business. Due to the rapidity and scope of this development, the Internet has become a vital resource for information sharing and a fundamental aspect of contemporary life. For business-critical tasks including banking, healthcare, education, government, human resources, smart cities, and grid systems, people and organizations use technology.

Although these duties are now easier to complete due to the need for quick access to information and advancements in information and communication technology, they nonetheless

present significant security problems that all parties involved—from individuals to governments—must address . As technology advances, people, societies, and organizations may be at danger from security vulnerabilities. People, businesses, and societies frequently use a variety of strategies to lessen these security issues, such as cybersecurity safeguards, legislative frameworks, and awareness and education campaigns that encourage moral behavior and deter malevolent intent. The defense of people, communities, institutions, systems, and technology against anomalous activity is known as cybersecurity. The preservation of the confidentiality, integrity, and availability (CIA) of computer resources that belong to one organization or are linked to the network of another is known as cybersecurity.

Since more than 61% of business and social contacts take place online, maintaining high security standards is crucial to enabling smooth, effective, and safe interactions. Data protection, privacy concerns, availability and dependability, and cyber security are some of the fundamental ideas to be taken into account in order to guarantee good security standards . As a result, cybersecurity has emerged as the most effective strategy for combating cybercrime and cyberattacks by ensuring secure social and interindustry contacts. One of the most important security concerns in the world today is making the internet safer and safeguarding internet users. Maintaining a safe online environment has significant ramifications for people, companies, and countries looking to enhance cyber security as the digital world keeps growing and integrating with nearly every part of our lives . The creation of new services, technology, and governmental regulations now heavily relies on internet security. Combating cybercrime necessitates a more thorough and secure strategy. This indicates that technological measures by themselves do not cause cybercrime. The ability of law enforcement organizations to efficiently investigate and prosecute cybercrime is vital . Strict cybersecurity regulations

are now in place in many nations, including Ethiopia, to stop information leaks.

Since almost every aspect of our life depends on technology to preserve social and economic stability and security, cybersecurity is essential in today's digital world. In sectors like healthcare and energy that depend on critical infrastructure, cyberattacks can have disastrous consequences, including monetary loss, damage to one's reputation, and even fatalities . Cybersecurity is used to safeguard critical infrastructures, ensure the confidentiality and integrity of sensitive data, and protect personal data in small businesses, government organizations, military units, healthcare facilities, educational institutions, energy suppliers, and transportation systems. Threat detection, network vulnerability discovery, and IT workload reduction are all possible with artificial intelligence (AI) and machine learning . Furthermore, many cybersecurity processes, including intrusion detection, malware analysis, and vulnerability assessment, can be automated with the help of machine learning and artificial intelligence (AI) . Security experts may be able to concentrate on more strategic work as a result.

Artificial intelligence (AI) and machine learning enable threat detection, network vulnerability identification, and IT workload reduction . Additionally, machine learning and artificial intelligence (AI) can be used to automate a number of cybersecurity procedures, such as vulnerability assessment, malware analysis, and intrusion detection . This could allow security professionals to focus on more important tasks.

By offering defense methods, intrusion detection systems, and encryption mechanisms, cyber security offers confidentiality, integrity, and reliability services for various domains . Because social contact and industry rely heavily on the internet, cyber security is primarily relevant in the following domains.

Future power systems with distributed intelligence, demand response, and renewable energy sources are expected to be more reliable and efficient thanks to the smart grid, the next generation of power systems that combines state-of-the-art processing and communication technology. The smart grid provides customers with faster and better services due to a lower response time delay, which makes it possible to implement energy-related solutions successfully . Because of the wide interconnection of electronic devices and communication networks in the electrical infrastructure, cybersecurity is a major risk for smart grids. Smart grids, which optimize electricity generation, distribution, and consumption through digital technology, have many advantages, including increased dependability and efficiency.

Communication networks for smart grid technology are essential for exchanging information in energy infrastructure. Confidentiality, integrity, and trust are the three primary cybersecurity objectives for the smart grid . Implementing encryption, secure communication protocols, authentication, and network monitoring for questionable activities can help achieve these cyber security objectives . Cybersecurity for smart grids guarantees rapid and dependable information access and utilization

Cybersecurity can reduce potential risks and hazards including denial of service (DoS), advanced persistent threats (APTs), and illegal access to data in order to preserve trust, confidentiality, and integrity in the smart grid system. In order to strengthen the Smart Grid's online security, cybersecurity highlights the potential negative consequences of cyberattacks, such as power outages, monetary losses, and safety hazards. Along with implementing state-of-the-art authentication and encryption methods, it is also advised to enhance threat intelligence and sharing, carry out regular security audits and assessments, and train stakeholders on cyber security best practices and

awareness.

The process of defending computer networks, systems, and data against online attacks that compromise the privacy, accuracy, and usability of information systems is known as cyber security. Because people and companies rely so heavily on digital technologies, cyber security is crucial to ensuring their safety.

Health facilities, financial institutions, smart cities, grid systems, government agencies, educational institutions, and the military are just a few of the application areas where security is relevant. Hackers, cybercriminals, state actors, terrorists, and insiders are some of the various threats that cyber security must contend with. Defensive AI and machine learning technologies, advanced cyberattacks, reinforcement learning-based cyberattacks, malware with AI capabilities, the vulnerability of IoT technology, cloud security concerns, and the use of cryptography are the difficulties facing the cyber security sector.

Ayurveda-Inspired Multi-Factor Authentication

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With the growing complexity of cybersecurity threats, traditional authentication methods are no longer sufficient. Multi-Factor Authentication (MFA) has become an essential tool for enhancing security by requiring multiple verification factors. This paper presents a unique approach to MFA, drawing inspiration from Ayurveda, an ancient Indian system of medicine. Ayurveda emphasizes personalized diagnostics, continuous monitoring, and adaptive treatments—principles that can be integrated into modern authentication methods. By incorporating biometric authentication, adaptive security measures, and contextual authentication, we propose a more resilient and user-friendly cybersecurity model.

Conventional authentication methods, such as passwords and One-Time Passwords (OTPs), are prone to phishing, brute-force attacks, and other security vulnerabilities. Ayurveda, which focuses on individual uniqueness, dynamic assessment, and holistic evaluation, provides an intriguing perspective on cybersecurity. This paper explores how Ayurvedic principles can be applied to enhance authentication systems, leading to a more dynamic and context-aware security framework.

Ayurveda and its Application in Authentication

Ayurveda classifies individuals based on the Tridosha system, which consists of three fundamental energies: Vata (Air & Space), Pitta (Fire & Water), and Kapha (Earth & Water). This personalized approach parallels biometric authentication, where each user has unique physiological and behavioral traits.

Moreover, Ayurveda uses techniques like Nadi Pariksha (pulse diagnosis) to assess an individual's health. Similarly, modern authentication systems can leverage behavioral biometrics, such as keystroke dynamics and gait analysis, to improve security.

1. Enhanced Biometric Authentication

Ayurveda considers physiological markers such as pulse patterns, eye movement, and tongue structure. Inspired by this, advanced MFA could incorporate:

- Iris movement tracking
 - Stress-based authentication
 - Dynamic heartbeat verification
- These biometrics add an additional security layer, making it difficult for attackers to impersonate users.

2. Adaptive Security Layers

Ayurvedic treatments are adjusted based on the changing state of an individual's dosha balance. Similarly, authentication systems should adapt to variations in user behavior.

- AI-driven continuous authentication can replace static logins.
- Security levels can be dynamically adjusted based on real-time risk analysis.

3. Holistic AI-Powered Security

Ayurveda considers environmental factors, emotions, and diet while diagnosing a person's health. Cybersecurity can adopt a similar approach by using context-aware authentication, including:

- Location-based verification (e.g., detecting familiar Wi-Fi networks)
- Behavioral analytics (e.g., analyzing typing speed and navigation habits)

The following steps can be implemented:

1. Biometric Data Collection

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Using sensors to capture biometric data such as pulse variations, facial expressions, and eye movement.

2.Machine Learning Integration

Developing AI models that analyze biometric and behavioral data to enhance security dynamically.

3.Context-Based Authentication

Implementing location-aware and device-based authentication to detect anomalies.

4.Dynamic Access Control

Establishing a risk-based authentication system that adjusts security levels in real time.

Some challenges

- Privacy Concerns – Securely handling sensitive biometric data.
- Sensor Accuracy – Ensuring high precision without causing user inconvenience.
- Computational Overhead – Optimizing AI-driven authentication for real-time performance.

The Journey of Knowledge from Traditional Era to Digital Era

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What is Knowledge? From where do you gain knowledge? What are the different dimensions in knowledge? To answer these questions, obviously maximum people would type the question on websites and AI apps. It has become common to gain maximum knowledge through secondary sources like magazines, books, AI apps – meta-AI, lensa, ChatGPT, Google Assistant, Amazon Alexa, DataBot Personal Assistant and the list can go on and on. The different types of knowledge such as explicit, tacit and implicit knowledge itself gives a broader perspective of different dimensions of knowledge. Theoretically to know about dimensions of knowledge I myself had googled and resulted with these – Epistemological Dimension, Ontological Dimension and Methodological Dimension. “It talks about sources, validity, relationship between knowledge and reality, and processes and techniques used to acquire and validate knowledge” - I am able to simplify these terms because of a subject being studied during my Bachelors and Masters of Education. Knowledge is a diversified broad concept and that is being paced from scientific knowledge, Philosophical Knowledge to Artificial Intelligence, virtual and Global Knowledge networks. Of course, “the knowledge that is to be gained through experience can never be replaced”.

A teacher gives assignment to her/his students and tells to submit the assignment within a specific time period, just analyze the way an assignment is completed by the 80's or 90's

kids and 2000 onwards born kids. This is the best example that shows us the commuting of traditional way of acquiring knowledge through books, magazines, journals to e- books, e- magazines, e- journals available just at a click. Majority of institutions and organizations are preferring Artificial Intelligence and various courses are online. Even vocational courses are conducted online. How far is it successful? It was found on YouTube by me about the results of learnings from virtual classes which was very hilarious but also found few sincere talented people benefiting from it. This is where I could see the difference in people who had gained knowledge from experienced Guru's directly and from experienced Guru's Virtual courses. Hence, through these videos I was able to know the benefits as well as the limitations of the dissemination of knowledge from over a period of time to present.

Every field requires research and innovation, consequently the curriculum includes research in every discipline – how effectively the importance and practical subject knowledge is created in the minds of students can be known through the mechanisms they have used while doing their research work. Again here, the means of their knowledge matters. A Debate can be raised on collection and analysis of data using digital mode is effective or collection and analysis of data personally is effective? - It has its own pros and cons. To list out there are many instances and examples that divulge about the means of gaining knowledge from past to present era that perceives the different dimensions of knowledge that is from oral traditions to written texts and from print media to digital platforms.

By reading this article you were able to know the evolution of disseminating the knowledge from past to present enlightening the different dimensions of knowledge specifically in educational field. Now here's a question put your knowledge to test. What would happen if a student asked their teacher in ancient Nalanda University to "Google" the answer to a question?

CYBER SECURITY AND ETHICAL HACKING: SAFEGUARDING THE DIGITAL WORLD

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In today's interconnected world, cybersecurity plays a vital role in protecting sensitive data, ensuring privacy, and preventing malicious attacks. With the rapid rise of cyber threats, the demand for ethical hackers—professionals who use their skills to identify and fix security vulnerabilities—has grown significantly.

Cybersecurity:

Cybersecurity refers to the practice of defending computers, servers, networks, and data from malicious attacks. It involves implementing security measures to protect against unauthorized access, data breaches, and cybercrimes. Cybersecurity has become a crucial aspect of personal and organizational safety.

Ethical Hacking

Ethical Hacking, also known as penetration testing or white-hat hacking, is the process of testing computer systems, networks, and applications to identify security flaws. Ethical Hackers operate with the organization's permission to improve security defenses.

Types of Ethical Hackers:

- 1.White Hat Hackers:** Authorized professionals who test systems for vulnerabilities.
- 2.Black Hat Hackers:** Criminal Hackers who exploit systems for illegal activities.
- 3.Gray Hat Hackers:** Individuals who hack without malicious intent but without proper authorization.

Importance of Ethical hacking in cybersecurity:

- **Identifying vulnerabilities:** Ethical hackers help the organizations discover security loopholes before criminals can exploit them. They provide security to the systems in such a way that criminals cannot hack the systems of organizations easily.
- **Preventing data breaches:** Ethical Hackers strengthens the security protocols to safeguard sensitive data.
- **Ensuring Compliance:** Ethical Hackers assists the organizations in meeting regulatory security standards.
- **Building Trust:** Enhances customer confidence by protecting personal information.
- **Encryption Decryption Testing:** Ethical hackers verify whether encryption algorithms used to protect data are secure or not.
- **Securing Databases:** They test for vulnerabilities such as SQL injection or data leakage, which would expose confidential information.
- **Wireless Network Security:** They conduct Wi-fi penetration tests to identify weak passwords or unprotected access points.
- **Testing Cloud Configuration:** Ethical Hackers detect misconfigurations, which are common early points for attackers,
- **Access Control Testing:** Ethical Hackers Verify if proper authentication and authorization mechanisms are in place.

CONCLUSION:

Ethical Hackers play an essential role in protecting organizations from cyber threats by identifying vulnerabilities, simulating attacks, strengthening security measures. Their skills and expertise help prevent data breaches, secure sensitive information, and ensure compliance with security regulations. As the digital landscape continues to evolve, the role of ethical hackers will remain indispensable in safeguarding the digital world.

Cybersecurity: Protecting the Digital World

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Introduction:

In today's digital era, cybersecurity is more crucial than ever. With increasing reliance on technology for communication, banking, business, and even healthcare, protecting sensitive data from cyber threats has become a priority for individuals, businesses, and governments. Cybersecurity encompasses a wide range of practices, technologies, and measures aimed at safeguarding computer systems, networks, and data from cyberattacks.

Understanding Cyber Threats:

Cyber threats are malicious activities that target digital systems, often with the intent to steal, alter, or destroy data. Some common types of cyber threats include:

- 1.Malware:** Malicious software such as viruses, worms, and trojans that infiltrate systems to steal or damage data.
- 2.Phishing:** Deceptive emails or messages designed to trick users into revealing personal information, such as passwords and financial details.
- 3.Ransomware:** A type of malware that locks or encrypts data, demanding a ransom for its release.
- 4.Denial-of-Service (DoS) Attacks:** Overloading a system or network with excessive traffic, rendering it inaccessible.
- 5.Man-in-the-Middle Attacks:** Intercepting communications between two parties to steal or manipulate data.
- 6.Zero-Day Exploits:** Attacks targeting vulnerabilities in software before developers can patch them.

Importance of Cybersecurity:

The impact of cyberattacks can be devastating, leading to financial losses, identity theft, data breaches, and reputational damage. Effective cybersecurity measures help:

- **Protect Personal Information:** Preventing identity theft and fraud.
- **Secure Businesses:** Safeguarding sensitive company data, intellectual property, and customer details.
- **Ensure National Security:** Preventing cyber espionage and critical infrastructure attacks.
- **Maintain Trust:** Ensuring secure online transactions and data integrity.

Best Practices for Cybersecurity:

1. Adopting strong cybersecurity practices is essential to mitigating risks. Some key measures include:

2.Strong Passwords: Use complex passwords and enable multi-factor authentication (MFA) for additional security.

3.Regular Software Updates: Keep operating systems, applications, and antivirus programs up to date to fix vulnerabilities.

4.Secure Networks: Use firewalls, VPNs, and encryption to protect network communications.

5.Beware of Phishing: Avoid clicking on suspicious links and verify the authenticity of emails and messages.

6.Backup Data: Regularly back up critical data to recover from ransomware attacks or system failures.

7.Access Control: Limit access to sensitive information and implement role-based permissions.

8.Cybersecurity Awareness: Educate employees and individuals about cyber threats and safe online practices.

The Future of Cybersecurity:

As technology evolves, so do cyber threats. Emerging technologies like artificial intelligence (AI), blockchain, and quantum computing will shape the future of cybersecurity. AI-driven threat detection, advanced encryption methods, and

decentralized security frameworks will play a crucial role in combating cyber threats.

Conclusion:

Cybersecurity is an ongoing battle that requires vigilance, innovation, and proactive measures. By staying informed and implementing robust security practices, individuals and organizations can protect themselves against cyber threats and contribute to a safer digital world. Investing in cybersecurity is not just an option—it is a necessity in the modern digital landscape.

Efficiency, Innovation, and Automation: The AI-IoT Revolution

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The Internet of Things (IoT) and artificial intelligence (AI) are transforming industries by fostering more intelligent, networked ecosystems. Automation, predictive analytics, and real-time data integration are all working together to increase production, reduce expenses, and improve efficiency across industries. Industries are paving the way for innovation-driven ecosystems that redefine contemporary operations as they adjust to this revolutionary synergy.

Introduction

In the age of digital transformation, Artificial Intelligence (AI) and the Internet of Things (IoT) are emerging as the cornerstones of innovation across industries. Together, these technologies are facilitating intelligent decision-making, automation, and smooth connectivity across a range of industries, including healthcare and manufacturing. This article examines how IoT and AI combine to build intelligent ecosystems that will influence industry trends in the future.

Understanding AI and IoT

- **Artificial Intelligence (AI):** Mimicking human intelligence, AI processes vast data sets to identify patterns and make autonomous decisions.
- **Internet of Things (IoT):** Interconnecting devices via sensors and networks to collect, share, and analyze real-time data.

Their collaboration fosters an environment where devices learn, adapt, and act autonomously, accelerating operational efficiency and innovation.

Applications Across Industries

1. Manufacturing: Building Smart Factories

IoT devices with AI capabilities are revolutionising industry through better product quality, reduced downtime, and enhanced predictive maintenance. Sensors assess the condition of the equipment and notify technicians to avoid malfunctions. AI-powered smart factories, for example, automate procedures and maximise resource allocation to ensure smooth operations.

2. Agriculture: Precision Farming for Better Yields

IoT sensors track crop health, soil moisture, and weather, and artificial intelligence (AI) evaluates data to suggest the best times to plant, water, and fertilise. By increasing productivity and reducing waste, precision farming promotes sustainability.

3. Healthcare: Transforming Patient Care

Healthcare is being redefined by wearable technology and IoT-powered smart diagnostics. Predictive analytics aids in early disease detection, while real-time monitoring permits prompt interventions. To improve individualised care, IoT-enabled systems, for example, monitor patient vitals and notify physicians of possible hazards.

4. Logistics: Fleet Optimization

By tracking fleet performance, optimising delivery routes, and guaranteeing on-time delivery, AI-driven IoT devices are transforming logistics. Supply chain activities are streamlined by real-time data analytics that lower fuel use and improve customer satisfaction.

Problems and Obstacles

IoT and AI convergence presents enormous opportunities,

however there are still obstacles to overcome:

1.Security and privacy: defending against cyberattacks on massive volumes of linked data.

2.Interoperability: Making sure that various IoT systems and devices integrate seamlessly.

3.Costs of Implementation: High upfront expenditure for software, hardware, and training.

4.Ethical Issues: Resolving AI bias and guaranteeing legal compliance.

Future Trends

1. Edge Computing and 5G Integration

Edge computing processes data locally, reducing latency and enhancing security. Coupled with 5G networks, it enables faster and more responsive IoT applications.

2. AI-Powered Autonomous Systems

Autonomous robots and self-driving vehicles are reshaping logistics, agriculture, and manufacturing. AI models continuously learn and adapt to IoT data, ensuring efficiency.

3. Blockchain for Security

Decentralized blockchain systems secure IoT data, preventing cyber threats and fostering trust.

4. Sustainable Solutions

Energy-efficient AI-IoT applications aim to reduce environmental impact, aligning with sustainability goals.

Conclusion

An era of intelligent and linked ecosystems is being ushered in by the combination of AI and IoT. Industries can fully utilise these technologies by tackling issues like high installation costs and security threats. AI-IoT-driven solutions will reshape industries as innovation advances, making them more intelligent, flexible, and effective.

Mathematics in Aerospace

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Mathematical problems in the aerospace industry extremely affect the development of various related fields, such as aerodynamics, astrodynamics, deep space exploration, space environment, manned space flight, space traffic management, etc.

The aerospace industry encloses the development of aircraft and spacecraft, including complex engineering challenges. Mathematics work as the foundational tool enabling engineers and scientists to model, simulate, and solve these challenges. From designing efficient wings to calculating optimal flight trajectories, mathematics is important in every aspect of aerospace engineering.

Aerospace engineering encompasses the design, development, and testing of aircraft and spacecraft. The area integrates various disciplines, with mathematics serving as a core component.

Mathematics allows engineers to model physical phenomena, analyse systems, optimize designs, and ensure operational reliability and safety. This article aims to explain the essential mathematical concepts and their applications in aerospace engineering.

Mathematics is major in understanding and optimizing the aerodynamic properties of aircraft.

Wing Design: Engineers use CFD to model airflow about different wing shapes, optimizing for lift and minimizing drag.

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Example - Designing a wing with a specific air foil shape that maximizes lift-to-drag ratio, improving fuel efficiency and performance

Structural Analysis Mathematics ensures that aircraft structures can withstand operational stresses without failure.

Stress Analysis: Using FEM, engineers predict how the aircraft's structure responds to various loads, ensuring it can withstand them without excessive deformation or failure.
Example - Analysing the stress distribution in an aircraft fuselage during flight to prevent structural failure.

Take-off and Landing: Newton's Second Law helps in calculating the thrust required for take-off and the braking forces needed for landing. For instance, during take-off, the thrust generated by the engines must overcome the drag and friction forces to accelerate the aircraft to lift-off speed. The relation $F=ma$ helps in determining the acceleration required and subsequently the thrust needed.

Launch and Orbit Insertion: When launching a spacecraft into orbit, Newton's Second Law helps determine the necessary thrust to overcome Earth's gravity (acceleration due to gravity) and achieve the desired orbital velocity. The spacecraft's mass m and the acceleration a (related to the rocket's thrust and gravity) are crucial parameters.

Reaction Control Systems (RCS): RCS thrusters on spacecraft use Newton's law to provide small adjustments in orientation and velocity. By applying force, these systems control the spacecraft's attitude and stabilize it during operations like docking manoeuvres or instrument pointing.

Atmospheric Entry: During re-entry into Earth's atmosphere, spacecraft experience deceleration due to atmospheric drag and

aerodynamic forces. Newton's law helps in calculating these forces and designing heat shields to manage the resulting heat and forces.

Air foil Design: The Naiver-Stokes equations help in understanding how air flows over wing surfaces. This analysis is essential for designing air foils that maximize lift and minimize drag.

Fuel Load Planning: Determining the optimal amount of fuel to carry for a given flight to minimize weight and maximize efficiency, while ensuring safety and regulatory compliance.

Fuel Distribution: Optimizing the distribution of fuel between different tanks in an aircraft to balance weight and maintain centre of gravity within safe limits.

Spacecraft Trajectory Optimization

Optimizing the path, a spacecraft takes involves solving complex mathematical problems to ensure efficient fuel use and mission success.

Hohmann Transfer: A fuel-efficient way to transfer between two orbits, calculated using Kepler's laws and optimization techniques. Example-Designing a transfer from Earth orbit to Mars orbit, minimizing fuel use while ensuring the spacecraft reaches its destination.

Lambert's Problem Lambert's Problem involves finding the orbit that connects two points in space in a given time frame.

Interplanetary Trajectories: Solving Lambert's Problem helps in planning missions to other planets by determining the best path and timing for launch. Example-Planning the trajectory for a probe to travel from Earth to Jupiter, ensuring it arrives at the right time for optimal scientific observations.

Finite Element Method (FEM)

FEM divides a complex structure into smaller elements, solving simplified equations for each element and combining the results.

Structural Analysis: FEM helps predict how parts of an aircraft or spacecraft deform under various loads.

Case Studies

1. Wing Design: Using FEM to optimize the wing structure for weight, strength, and aerodynamic performance. This involves detailed stress analysis, buckling analysis, and aero elastic studies.

2. Engine Components: Thermal and structural analysis of turbine blades to ensure they can withstand high temperatures and stresses over long operational periods.

3. Spacecraft: Analysis of thermal stresses and structural integrity of spacecraft components, such as the heat shield during re-entry.

4. Landing Gear: Dynamic and stress analysis of landing gear to ensure safe operation during take-off, landing, and taxiing.

Example - To analyse stress distribution in an aircraft wing, engineers model the wing as a collection of finite elements and solve for stresses and strains in each element

In conclusion, mathematics is not merely a tool but a cornerstone of aerospace engineering. It underpins every aspect of the field, from conceptual design and testing to the operational dynamics of flight and space exploration. The continual advancement of aerospace technology is deeply rooted in the development and application of sophisticated mathematical techniques. As aerospace engineers tackle increasingly complex challenges, the role of mathematics will only grow, driving innovation and ensuring the continued success and safety of aerospace endeavours.

Navigating the Cyber Threat Landscape: The Role of Artificial Intelligence

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The purpose of cyber security is to protect data and information systems (including networking, computing systems, cloud services, and software). Using tools like firewalls, virus protection, and other software to safeguard computer networks and personal data is essential, but it is insufficient. Cyber-security must be considered by Internet service providers (ISPs) and information technology (IT). We must be mindful of the various threats that are there in the online environment. Strengthening cyber security and protecting important data is essential to the country's security, stability, and well-being. The growing issue of online crime is the first thing that comes to mind when we think of cyber security.

Governments and corporations are taking a number of actions to fight cybercrime. The frequency of cybercrime with smartphones and private computers emphasizes the importance of staying up to date with recent global trends in cybercrime. Despite various protections, cyber security continues to be a major issue for many individuals. With the aid of artificial intelligence, this article focuses on the latest developments in cyber security strategies and trends.

Cyber security using AI

By enabling the creation of Intelligent Agents to successfully handle certain security issues, artificial intelligence (AI) has shown itself to be a vital tool in the fight against cybersecurity

issues. Through its ability to observe, learn, and make decisions, an intelligent agent—whether it be software or hardware—is made to maximize the likelihood of achieving a specific goal. These intelligent agents are able to spot flaws in intricate code structures, spot anomalies in user login habits, and even spot new malware strains that avoid detection by traditional means.

Finding flaws or weaknesses that an attacker could exploit is frequently the first line of defence. AI helps improve the accuracy of source code scanning, which reduces false positives and allows engineers to find security flaws before putting apps into production.

Responding to threats is another area in which AI helps security workers. The security incident response team receives contextual information and threat intelligence from advanced AI technologies. The team's entire incident response capabilities are improved by this context, which enables them to react to threats effectively and efficiently.

AI-Based Cybersecurity Techniques

- Machine learning (ML) algorithms: supervised, unsupervised, and reinforcement learning techniques are frequently employed to categorize data and spot possible threats.
- Natural Language Processing (NLP): Assists in separating dangerous information, spam, and phishing emails from authentic correspondence.

Machine learning (ML) algorithms

Machine learning continuously scans the network's activity for irregularities in order to identify risks. To find important instances, machine learning engines analyse vast volumes of data almost instantly. These methods make it possible to identify policy infractions, unknown malware, and insider risks. At machine speed, machine learning can identify patterns and anticipate dangers in large data sets. Cyber teams can quickly identify threats and identify circumstances that require more in-

depth human investigation by automating the analysis.

In order to help stop users from visiting harmful websites, machine learning can identify "bad neighbourhoods" on the internet. Machine learning automatically detects attack infrastructures set up for emerging and existing threats by analysing Internet behaviour.

Several machine learning methods, each with unique uses and advantages, are frequently employed to forecast and identify cybersecurity threats:

1.Decision Tree

An essential kind of algorithm for machine learning and predictive modelling is the decision tree. Modern variants of the traditional decision tree algorithms, such as random forest, are among the most potent methods now in use. These algorithms have been around for decades.

This algorithm is known as "decision trees" in the classical sense, however in some platforms, such as R, it is called CART.

The name Classification and Regression Trees, or CART for short, was coined by Leo Breiman to describe Decision Tree algorithms that can be used to problems involving regression predictive modelling or classification. Important algorithms including bagged decision trees, random forests, and boosted decision trees are built on top of the CART algorithm.

A binary tree serves as the CART model's representation. This is a basic binary tree based on data structures and algorithms. A single input variable and its split point are represented by each root node. A prediction is made using an output variable found in the tree's leaf nodes.

In order to traverse the tree given a new input, the particular input is evaluated beginning at the tree's root node.

2. Random Forest

One of the most well-liked and potent machine learning algorithms is Random Forest. This kind of ensemble machine learning approach is known as bagging or bootstrap aggregation.

We can utilize several trees with a random sample of features selected as the split to enhance decision tree performance. An ensemble technique called Random Forest builds several decision trees during training and aggregates their results to get a final forecast. It is predicated on the idea that the combined findings of several models are more trustworthy than those of a single model, or the "wisdom of the crowd."

Step 1: Data sampling, or bootstrapping

Random Forest uses bootstrapping, a sampling technique in which samples are taken at random with replacement, to generate multiple subsets of the training data. Because each tree in the forest is trained on a slightly different dataset, variability is introduced. This implies that each subset may contain duplicates of certain data points and may exclude others.

Step 2: Using Random Feature Selection to Build Decision Trees

Random Forest does not utilize every feature for every decision tree at every split. Rather, it chooses a subset of features at each node at random. This promotes diversity in the forest by lowering correlation between the trees and guaranteeing that each tree concentrates on a distinct area of the dataset. Since no pruning is done, each tree grows to its full potential, making them extremely deep and specific to the subset of data they are trained on.

Step 3: Combining Outcomes

After every tree has been trained, Random Forest aggregates the outcomes to produce a final forecast. To determine the final class label, a majority vote from each tree is required. For

regression, the final result is obtained by averaging the predictions made by each tree.

Using NLP in Cybersecurity

In the field of cybersecurity, natural language processing has become a potent technology that allows machines to comprehend and analyse large volumes of textual data, improving overall security posture. NLP can be used to support a variety of cybersecurity models, such as,

Threat Intelligence and Monitoring

NLP is essential to the collection and tracking of threat intelligence. NLP can find trends, extract pertinent information, and reveal possible dangers by analysing data from a variety of sources, including social media, forums, news articles, and chats on the dark web. Security teams may keep ahead of new cyber threats by using techniques like named entity identification, sentiment analysis, and subject modelling to filter and prioritize threat intelligence.

Identification of Social Engineering

In order to deceive victims, phishing and social engineering assaults frequently use misleading wording. To find suspect trends, odd demands, or copied writing styles, natural language processing (NLP) can analyse the language used in emails, websites, and other interactions. NLP can strengthen an organization's defences against these kinds of attacks by utilizing methods like text classification, sentiment analysis, and anomaly detection.

Analysis of Threats and Malware

To find potential vulnerabilities, malicious code, or indicators of compromise (IOCs), NLP can be used to analyse malware samples, code repositories, and threat reports. Understanding the purpose and behaviour of malware can be facilitated by methods like as code creation, semantic parsing, and natural

language inference. This allows for more efficient threat analysis and mitigation tactics.

Analysis of Logs and Incidents

Unstructured language data in security logs and event reports frequently conceals important information. Logs and reports can be analysed by NLP, which can extract crucial information and identify odd trends that might indicate security problems. Methods such as entity extraction, relation extraction, and text summarization help speed up incident response and the forensic investigation process

Pi Unleashed

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Most people remember Pi from geometry class as a weird number that begins at 3.14 and goes on indefinitely. However, this mathematical constant has far-reaching applications outside of the classroom, playing an important part in everything from smartphones to the satellites orbiting above. Geometric approaches use various shapes and constructs to approximate the value of Pi. Archimedes produced one of the earliest geometric procedures, inscribing and circumscribing polygons around a circle to approximate Pi. By increasing the number of sides of the polygons, he was able to achieve increasingly precise Pi estimates.

Pi has a huge impact. Pi is celebrated by mathematicians worldwide because of physicist Larry Shaw, who established International Pi Day on March 14 (3.14), in 1988. This irrational number influences our everyday lives and is present everywhere, from scientists measuring light waves to engineers assessing building integrity. For thousands of years, the idea of Pi has captivated academics. This unique ratio was recognized and frequently approximated by ancient civilizations ranging from Egypt to China. Pi can now be calculated with previously unheard-of accuracy because of computer science, opening the door to intricate space exploration and precise architecture. In the '90s, a flaw in a computer chip was discovered when a mathematician noticed incorrect numbers while performing number theory computations on the chip. This flaw was fixed after the mathematician informed the computer company.

The constant Pi appears in many natural phenomena, including the way light waves ripple through water droplets to make rainbows and the spiral development of galaxies in space. The number can be seen in the circular patterns of ripples left by a raindrop on a calm pond, as well as in the mathematical descriptions of natural vortices ranging from microscopic whirlpools to large air systems. Engineers utilize Pi to create precision instruments such as accelerometers and gyroscopes that require accurate circular measurements. Pi is used in fluid dynamics to determine how liquids and gases flow through pipelines and around barriers, which allows for everything from efficient fuel systems to wind tunnel testing. When engineers and scientists study waves whether in water, sound, or light Pi is essential to understanding their behavior. This relationship determines everything from acoustic design in concert halls to signal processing in Wi-Fi networks. Even the way light waves propagate through space follows patterns governed by Pi.

Pi is used extensively in space exploration. Pi is used by scientists and engineers to compute the volume of a sphere when analyzing planets, and orbital mechanics relies on it to estimate spacecraft trajectories. The numerical theory underpinning Pi is even used to compute the positions of heavenly bodies.

Architects and construction engineers typically employ Pi while building curved constructions. The circumference of a circle or circular building cannot be stated simply as a fraction of its

diameter; here is where Pi comes in. Whether estimating the materials required for a dome or establishing the load distribution in an arch, Pi is indispensable. Modern architects rely on Pi to design energy-efficient curved surfaces and geometric facades that optimize natural lighting and airflow, just as ancient mathematicians used it to build the pyramids and modern engineers to design skyscrapers. The challenge of

achieving the required accuracy in these calculations has driven both mathematical innovation and engineering precision.

Even athletics uses Pi. Sports equipment manufacturers use Pi to calculate the best curvature of a basketball or the aerodynamics of a tennis racket. The circumference of any ball, from soccer to baseball, is described using Pi. To analyze sports performance, training facilities utilize computer models that employ Pi. Pi helps sportsmen and coaches optimize performance by estimating the optimum trajectory for a shot or constructing the appropriate curve of a racetrack. Pi is used by swimming pool builders to compute volume and water flow, as well as track designers to establish appropriate curve radii for peak running speeds.

As we travel further into space, the precision of our computations becomes more important. Pi's infinite digits offer the precision required for such sophisticated calculations. Mathematicians continue to calculate additional Pi digits, despite the fact that most space applications require only a few. NASA engineers frequently utilize Pi to calculate everything from rocket design to planetary exploration. Pi's effect now extends far beyond geometry class. Whether you're using your smartphone's GPS, watching a satellite TV broadcast, or enjoying a curved building, you're seeing Pi at work. Pi continues to demonstrate its utility in anything from cutting-edge research facilities to common electronics.

In Environmental Science, π ($\pi \approx 3.1416$) is commonly utilized in quantifying natural formations including tree rings, lakes, and planetary bodies. It aids in estimating forest extent, tree canopies, and water bodies for conservation purposes. In climate research, π is used in equations that describe atmospheric patterns, ocean waves, and planetary movements to aid in climate prediction. It also contributes to the design of renewable energy systems like as wind turbines, solar panels,

and hydropower plants. Thus, π is vital in understanding natural processes and implementing sustainable environmental solutions.

Despite substantial research and understanding of π , some unsolved puzzles and riddles continue to fascinate mathematicians and scientists. One such puzzle is the distribution of π 's digits. While the digits of π appear to be randomly and uniformly distributed, there is no definitive proof to support this observation. As a result, research on the distribution of π 's digits continues to be active. The question of whether π is a normal number remains unanswered. A number is considered normal if its digits, in any base, are uniformly distributed and independent of one another. While it is widely assumed that π is a normal number, no proof has been established to support this hypothesis.

As we continue to investigate the secrets of π , fresh discoveries and breakthroughs are expected to emerge, adding to our grasp of this enigmatic constant. The emergence of new mathematical methodologies, computational algorithms, and scientific applications will definitely contribute to π 's enduring appeal.

Polymer-Ferrite Nanocomposites for sensing applications: A Brief Review

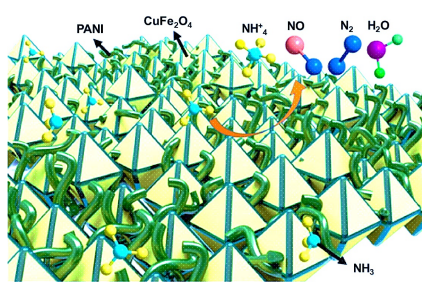
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Introduction

Polymer nanocomposites have gained much attention with a wide range of sensing applications due to unique combination of magnetic and electrical properties, alongside the versatility and ease of processing of polymers [1]. Ferrites were found unequal magnetic properties to take hold of impulsive magnetization, which can be tailored for many applications which can be combined with different elements [2]. Ferrites were fused into polymers to produce nanocomposites that exhibits in enhancing electronic, electrical, thermal, and mechanical characteristics [3]. This material explores the properties, and applications of polymer-ferrite nanocomposites in the field of sensors, which particularly focus on magnetic, piezoelectric, and environmental sensing [4]. The challenges, future trends, and potential advancements in the development of these materials.

Polymer-based nanocomposites have gained more attention in recent years due to their less weight, plasticity, and tuneable properties [4]. The combination of organic and inorganic materials acquires an enhanced functionalities that can make nanocomposites for usage in sensing technologies which is depicted in figure 1 [4]. Ferrites are ceramic compounds consisting of iron oxides with other metal oxides, that can enhance the sensing abilities of polymer matrices [5]. The incorporation of ferrite nanoparticles into polymers matrix, which results in nanocomposites that has combined properties with enhanced sensitivity and mechanical strength of ferrites

with the flexibility and ease of processing characteristics to polymers [6]. Polymer-ferrite nanocomposites are used for sensing applications is rapidly expanding, driven by their potential in various industries, including environmental monitoring, biomedical devices, automotive systems, and industrial automation [7]. This review aims to provide an overview of the recent advancements in polymer-ferrite nanocomposites, focusing on their use in magnetic, piezoelectric, and biological sensors.



*Figure 1: Schematic representation of ammonia gas sensing
Polymer-ferrite nanocomposites*

Synthesis of Polymer-Ferrite Nanocomposites

The synthesis of polymer-ferrite nanocomposites typically involves the dispersion of ferrite particles within a polymer matrix, which can be achieved through various processing methods such as melt blending, solution casting, and in situ polymerization techniques.

1.Melt Blending: In melt blending, ferrite particles are mixed with the molten polymer under controlled conditions. This method is direct and cost-effective but ferrites are unevenly dispersed and it cannot be controlled [8].

2.Solution Casting: In this method, the polymers is dissolved in a solvent, and ferrite particles were dispersed in the polymer solution. The mixture is then cast into thin films or other desired shapes. This technique allows better control over the dispersion of the ferrites but requires the use of solvents, which may pose

environmental concerns [9].

3.In Situ Polymerization: In situ polymerization involves the polymerization of monomers in the presence of ferrite nanoparticles, leading to the formation of the composite. This method offers a high degree of control over the morphology and distribution of ferrite particles [10].

Properties of Polymer-Ferrite Nanocomposites

Polymer-ferrite nanocomposites exhibit a range of enhanced properties due to the synergistic interaction between the polymer matrix and ferrite particles:

1.Magnetic Properties: Ferrites are magnetic materials, and their inclusion in a polymer matrix imparts magnetic properties to the composite. The type, size, and concentration of ferrite particles in the composite determine the magnetic characteristics, such as coercivity, saturation magnetization, and magnetic permeability. These properties are crucial for applications in magnetic sensors, inductive sensors, and electromagnetic interference (EMI) shielding.

2.Electrical Properties: The addition of ferrite particles can modify the electrical conductivity of the composite. This is particularly useful for applications such as piezoresistive sensors, where changes in resistance are used to detect mechanical deformations. The conductivity can be tuned by varying the ferrite content and polymer type.

3.Mechanical Properties: Polymer-ferrite nanocomposites often exhibit improved mechanical strength, stiffness, and durability compared to pure polymers. The dispersion of ferrite particles within the polymer matrix can reinforce the material, making it suitable for applications in harsh environments.

4.Thermal Stability: Ferrites typically exhibit good thermal stability, and their inclusion in the polymer matrix can improve the thermal performance of the composite. This is particularly advantageous for high-temperature sensing applications, such as temperature sensors or sensors used in automotive and industrial environments.

Applications in Sensing Technologies

The unique properties of polymer-ferrite nanocomposites make them suitable for a wide variety of sensing applications. Some of the key areas of application include:

1.Magnetic Sensing

Magnetic sensors, such as magnetic field sensors, magneto resistive sensors, and inductive sensors, are among the most prominent applications of polymer-ferrite nanocomposites. The magnetic properties of ferrites allow for the detection of external magnetic fields, making these nanocomposites ideal for applications in position sensing, current sensing, and proximity detection. Polymer-ferrite nanocomposites can be used to create flexible magnetic sensors that are lightweight and can be integrated into wearable devices, offering significant advantages in terms of portability and ease of use.

2.Piezoelectric Sensing

Polymer-ferrite nanocomposites can also be designed to exhibit piezoelectric properties, where the composite generates an electrical charge in response to mechanical stress. These materials are widely used in sensors for pressure, vibration, and strain measurements. By adjusting the composition and size of the ferrite particles, the piezoelectric response can be tuned for specific applications. For example, these nanocomposites can be used in structural health monitoring systems, wearable sensors, and environmental monitoring devices.

3.Environmental Sensing

The integration of ferrites into polymer matrices can enhance the sensing of environmental parameters such as humidity, temperature, and gases. Ferrite-based polymer nanocomposites can be tailored for specific sensing applications, including gas detection (e.g., for toxic gases or pollutants), humidity sensing, and temperature measurement. These sensors can be deployed in industrial settings, smart homes, and wearable environmental monitoring devices.

4.Biomedical Sensing

Polymer-ferrite nanocomposites have shown promise in

biomedical applications, such as biosensors and medical diagnostics. The magnetic properties of ferrites enable the detection of biomolecules through magnetic resonance imaging (MRI), magneto-optical sensing, and other magnetic-based detection techniques. Additionally, the biocompatibility and flexibility of polymers make these nanocomposites suitable for wearable health monitoring devices and implantable sensors.

5. Electromagnetic Interference (EMI) Shielding

Ferrite-based polymer nanocomposites are also being explored for use in electromagnetic interference (EMI) shielding. The magnetic properties of ferrites help in attenuating electromagnetic waves, making these nanocomposites useful in protecting sensitive electronic devices from interference. These materials can be used in applications ranging from mobile phones to military electronics.

Challenges and Future Directions

Polymer-ferrite nanocomposites hold great promise for sensing applications, but several challenges must be addressed. One key difficulty is achieving uniform dispersion of ferrite particles within the polymer matrix, particularly at high conductivity. Poor dispersion can lead to reduced performance and inconsistencies in sensing capabilities. Additionally, the interaction between the polymer matrix and ferrite particles plays a crucial role in ensuring optimal mechanical, electrical, and thermal properties. Developing effective surface treatments for ferrite particles to enhance the interface with the polymer remains an active area of research. The synthesis methods for these nanocomposites, such as solution casting or in situ polymerization, often involve complex processes that may not be easily scalable for industrial production. Environmental concerns also arise from the use of certain solvents during processing, highlighting the need for more sustainable, green synthesis methods for broader adoption. Finally, the cost of ferrites—particularly specialized or high-quality types—can be prohibitive, making it essential to explore cost-effective

alternatives or optimize ferrite loading to improve affordability.

Future Scope

Future research in polymer-ferrite nanocomposites for sensing applications is likely to focus on:

Smart and Multifunctional Sensors: Combining multiple sensing modalities in a single composite (e.g., magnetic, piezoelectric, and environmental sensing) could lead to more advanced smart sensor systems.

Nanostructured Ferrites: The development of nanostructured ferrite particles, which have a higher surface area and enhanced properties, could significantly improve the performance of polymer-ferrite nanocomposites.

Sustainability and Green Processing: Developing environmentally friendly methods for synthesizing polymer-ferrite nanocomposites, such as using renewable polymers or non-toxic ferrite precursors, will be crucial for future applications.

Wearable and Flexible Sensors: The increasing demand for wearable sensors and flexible electronics will drive the development of polymer-ferrite nanocomposites with enhanced flexibility, durability, and sensitivity.

Conclusion

Polymer-ferrite nanocomposites represent a promising class of materials for advanced sensing applications. Their ability to combine the mechanical flexibility of polymers with the magnetic and electrical properties of ferrites opens a wide range of potentials in healthcare industries and environment. While challenges remain in terms of dispersion, interface optimization, and scalability were ongoing for research and technological advancements hold great promise to realize the full potential of these nanocomposites in sensing technologies.

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“Revolutionizing Research Paper Writing: The Role of AI Tools”

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Introduction:

Writing a research paper is a tough job that needs careful planning deep study, and precise work. Researchers often struggle to find good sources, keep their work original, handle citations, and present their findings well. In recent years, AI has changed things by offering tools that make many parts of academic writing better and easier.

AI tools help scholars focus more on new ideas and critical thinking by doing citations making the language better, and spotting copied work, among other things. This article looks at the different AI tools for writing research papers that are out there talking about what they do, what they're for, and how they make academic work more productive. By using these tools the right way, researchers can write high-quality important papers more.

Different tools help researchers write papers:

1.Literature Review Tools

Purpose: To find, organize, and combine previous studies.

- **Zotero:** A free app to gather, sort, and reference research. It pulls info from websites and PDFs on its own building a list of sources.
- **Mendeley:** Gives teams a way to work together online letting them share and comment on papers.
- **Connected Papers:** Makes a picture of how research articles link to each other based on one main paper making it simpler

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to spot useful studies.

- **Research Rabbit:** A tool that finds links between papers and suggests more things to read.

2. Writing and Content Generation Tools

Purpose: To help draft, summarize, or polish research content.

- **ChatGPT (OpenAI):** Great for coming up with topics creating first drafts, or making complex ideas easier to understand.
- **QuillBot:** Helps reword or condense big chunks of text, which comes in handy when fine-tuning literature reviews or abstracts.
- **Jasper:** Made to produce professional-grade content with tips to boost clarity and organization.

3. Enhancing Grammar, Style, and Tone

Purpose: Refining language to meet academic standards.

- **Grammarly:** This tool catches grammar mistakes, suggests clearer sentence structures, and helps adjust the tone to suit academic writing.
- **ProWritingAid:** Offers detailed feedback on sentence flow, word choice, and readability, making your writing polished and professional.
- **Hemingway Editor:** Highlights long or complex sentences, helping you write clearly and concisely.

4. Ensuring Originality with Plagiarism Detection

Purpose: Ensuring originality and academic integrity.

- **Turnitin:** Compares your work with a vast database of papers, websites, and articles to detect potential plagiarism.
- **Grammarly Premium:** Its built-in plagiarism checker scans internet sources and academic databases for overlapping content.
- **Copyscape:** A quick, easy tool to ensure your content is unique, especially when publishing online.

5. Data Analysis and Visualization

Purpose: Handling and presenting research data.

- **Tableau:** Turns raw data into interactive visuals like charts and dashboards, making it easier to interpret and share findings.
- **MATLAB:** Ideal for complex computations and data modeling, especially in technical fields.
- **Orange:** Simplifies data mining and visualization, particularly useful for analyzing large datasets.

6. Citation and Reference Management

Purpose: Automating bibliography and in-text citations.

- **EndNote:** Lets you organize references, insert citations as you write, and automatically create bibliographies in the required format.
- **CiteThisForMe:** Quickly generates citations in different styles, saving time when referencing.
- **RefWorks:** Great for collaborative research, allowing team members to share and organize references seamlessly.

7. Language Translation and Accessibility

Purpose: Making research multilingual and globally accessible.

- **DeepL:** Known for its high-quality translations, especially for academic and technical content.
- **Google Translate:** A fast option for translating content, continually improving in accuracy for academic purposes.
- **Papago:** Focused on Asian languages, making it especially helpful for cross-cultural collaborations.

8. Text Summarization and Abstract Writing

Purpose: Summarizing lengthy documents or creating concise abstracts.

- **Scholarcy:** Pulls out key points and sections from lengthy papers, helping you get to the heart of the content faster.
- **Resoomer:** Condenses long articles into brief summaries, perfect for quickly skimming through multiple papers.
- **SciSpace Copilot:** Not only summarizes content but also

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explains complex terms, making it easier to understand challenging topics.

9. Formatting and Structuring

Purpose: Structuring research papers to meet journal or institutional standards.

- **Overleaf:** A collaborative platform based on LaTeX, widely used for formatting papers in science and engineering.
- **Typeset:** Provides ready-made templates for specific journals, making it simple to meet formatting requirements.
- **Manuscripts.io:** Helps structure, format, and export your paper with ease.

10. Proofreading and Final Review

Purpose: Enhancing the quality of the final draft.

- **PaperRater:** Checks grammar and style while providing insights into the originality of your content.
- **Slick Write:** Points out stylistic errors and helps improve readability.
- **AutoCrit:** Ensures consistency in tone, sentence structure, and overall writing quality.

11. Generating and Organizing Ideas

Purpose: Generating research ideas and organizing thoughts.

- **MindMeister:** A mind-mapping tool that visually organizes your ideas, helping you outline your research effectively.
- **IdeaNote:** Facilitates brainstorming and collaboration to refine research topics.
- **Scapple:** A free-form tool for connecting and organizing thoughts, fostering creativity.

How AI Tools Benefit Researchers

1.Save Time: Automating tasks like citations, formatting, and visualizations allows researchers to focus on analysis and writing.

2.Increase Accuracy: Tools like Grammarly and Turnitin

ensure your work is clear, error-free, and original.

3.Encourage Collaboration: Platforms like Overleaf make it easier for teams to work together, even remotely.

4.Improve Accessibility: Translation tools help researchers share their findings with a global audience.

Conclusion

AI tools are transforming the way researchers approach their work. From refining language and detecting plagiarism to formatting papers and visualizing data, these tools help streamline every step of the process. However, it's essential to use them responsibly to uphold academic integrity. By incorporating AI effectively, researchers can produce high-quality, impactful work while saving time and effort.

“Breathless: The Scientific Race to Defeat a Deadly Virus” – A Narrative of Science in Action

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David Quammen’s *Breathless* demonstrates how storytelling can make scientific topics easy to understand and engaging. The book focuses on the scientists trying to understand SARS-CoV-2, the virus responsible for COVID-19, but it is more than just a report on lab experiments. It takes readers into the heart of the scientific process, making the virus itself the main character of the story. As Quammen guides us through the global effort to solve the mystery of this virus, he shows how storytelling plays a crucial role in science communication.

At the core of *Breathless* is the search for the origins of SARS-CoV-2. Quammen structures the narrative like a detective story, with each chapter presenting a new scientific puzzle—whether it’s identifying where the virus came from, tracking its genetic mutations, or learning how it spreads. The journey spans remote caves in China to cutting-edge laboratories worldwide. Through this exploration, Quammen introduces us to genomic epidemiologists, evolutionary virologists, and other experts working tirelessly across borders to make sense of the virus.

The book includes insights from interviews with 95 scientists, providing a rare look at the process of scientific discovery. Quammen doesn't only focus on successes—he also highlights the frustrations, uncertainties, and setbacks these researchers faced. This approach offers readers a deeper understanding of pandemic science, emphasizing that scientific progress is rarely straightforward. Hypotheses must be tested and retested, and answers are not always clear, especially when dealing with a rapidly changing virus.

One of the most interesting aspects of *Breathless* is how it shows the importance of genomics in fighting COVID-19. Quammen explains how scientists used genomic sequencing to track mutations and understand how the virus spread. For example, he describes how new variants like Delta and Omicron affected the course of the pandemic, challenging public health systems around the world. Quammen simplifies this complex topic by framing it as a narrative of pursuit and discovery.

He takes readers inside the world of scientists, sharing their struggles and triumphs as they traced the virus's journey from bats to humans, possibly through an intermediate animal host. This storytelling approach not only makes technical concepts easier to grasp but also highlights the emotional and intellectual commitment of the researchers involved.

One of the book's strengths is how it portrays the human side of science. Quammen shows how scientists collaborated across countries, sharing data and insights in real-time to combat the crisis. However, the narrative also includes the conflicts and challenges that arose—whether among governments, public health agencies, or even between researchers. This honest portrayal offers readers a realistic view of the political and personal dynamics that complicated the global response to the pandemic.

Through these elements, Quammen breaks down the myth of science as a purely objective and linear process. Instead, he shows it as a human endeavor, shaped by ambition, curiosity, and sometimes rivalry. This adds depth to the narrative, reminding readers that science is not just about finding solutions but also about navigating uncertainties and challenges.

Quammen also touches on the difficulties of communicating science during a pandemic. Even well-meaning scientists and public health officials struggled to explain their findings clearly to the public. For example, the confusion around airborne transmission and changing mask guidelines reflected the evolving nature of scientific understanding. Quammen emphasizes that science is a process of gradual learning, where early conclusions often change with new data. However, this process is hard to communicate in an era of instant information and polarized opinions.

In this way, *Breathless* is not just a book about the virus but also a commentary on the importance of science communication. Quammen's ability to tell an engaging story while respecting the complexity of scientific work sets an example for effective communication. His narrative reminds readers that uncertainty is not a failure of science but a natural part of the process.

A unique aspect of *Breathless* is how Quammen makes the virus itself the protagonist. By placing SARS-CoV-2 at the center of the story, he shifts the focus from human-centered responses to the virus's own evolution. This approach offers readers a fresh perspective, emphasizing the virus's ability to adapt and survive. Quammen encourages readers to think about the virus not just as an enemy but as a complex part of nature, shaped by evolutionary pressures.

This shift in perspective challenges the simple idea of viruses as enemies and invites readers to reflect on the relationship

between humans and the natural world. Through this narrative, Quammen transforms the scientific story into a gripping account of evolution and survival.

Quammen's storytelling bridges the gap between science and public understanding. His narrative offers an insider's view of the scientific process, showing both the triumphs and the challenges scientists faced. By humanizing the science behind pandemics while maintaining respect for its complexity, *Breathless* serves as an excellent example of how to communicate science effectively.

The book also highlights the importance of interdisciplinary collaboration in solving global problems. Just as scientists from different fields came together to understand SARS-CoV-2, Quammen's narrative brings together elements of biology, epidemiology, politics, and communication to offer a well-rounded view of the pandemic.

In doing so, *Breathless* demonstrates that science and storytelling are not opposites but complementary tools. Quammen's ability to weave these elements into a cohesive narrative shows how stories can inspire, inform, and engage the public, making complex scientific ideas more accessible to everyone.

THE ROLE OF ARTIFICIAL INTELLIGENCE IN MODERN CYBERSECURITY

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INTRODUCTION

Let us understand what is AI in cybersecurity, Threats, key uses and Applications of cybersecurity.

Why is cybersecurity important today?

Cybersecurity is important today because it protects our personal data, money, and systems from growing online threats like hacking and fraud. As we use technology more, strong cybersecurity keeps us safe and secure in the digital world.

Growing cyber threats:

These are some common cybersecurity threats

1. Phishing
2. Ransomware
3. Malware
4. Data Breaches
5. Denial of Service (DoS/DDoS) Attacks
6. Man-in-the-Middle (MitM) Attacks
7. Insider Threats
8. Zero-Day Exploits
9. SQL Injection
10. Social Engineering

What is AI in cybersecurity?

As cyberattacks grow more sophisticated, traditional security measures are struggling to keep up. Artificial Intelligence (AI) is stepping in as a game-changer, revolutionizing how we protect sensitive information. This article explores how AI is

enhancing cybersecurity by improving threat detection, defense, and even predicting future attacks, from machine learning to deep neural networks.

KEY USES OF AI IN CYBERSECURITY

Here are the key uses of AI in cybersecurity:

- 1. Threat Detection**-AI identifies unusual behavior or patterns that may signal a cyberattack.
- 2. Real-Time Monitoring**-Continuously watches systems to detect and respond to threats instantly.
- 3. Automated Responses**-AI can quickly block threats or isolate infected systems without human help.
- 4. Phishing Detection**-Detects fake emails or websites designed to steal information.
- 5. Malware Analysis**-Identifies and analyzes new or unknown malware faster than humans.
- 6. Predictive Security**-Uses past data to predict and prevent future cyberattacks.

BENEFITS OF AI IN CYBERSECURITY

1. Enhanced Threat Detection

- **Real-time Analysis:** AI-powered systems can analyze vast amounts of data in real time, identifying suspicious patterns and potential threats faster than humans. This includes recognizing anomalies in network traffic, system behavior, or user activity.
- **Advanced Pattern Recognition:** Machine learning algorithms can detect patterns in data that may indicate emerging threats, such as zero-day attacks or advanced persistent threats (APTs), even before traditional signature-based systems can recognize them.

2. Predictive Capabilities

- **Threat Prediction:** AI can analyze historical attack data and predict potential future threats. By identifying patterns and trends, AI helps cybersecurity teams anticipate new attack strategies and proactively strengthen defenses.

- **Vulnerability Management:** AI can also prioritize vulnerabilities in a system based on threat intelligence, helping organizations patch or fix the most critical issues before they can be exploited.

3. Improved Malware Detection

- **Behavioral Analysis:** Instead of relying solely on signature-based detection (which identifies known malware), AI uses behavioral analysis to detect malware by how it behaves on a system. This allows AI to identify new, unknown malware strains (zero-day malware) that haven't been cataloged yet.
- **Advanced Malware Classification:** AI models can classify malware based on its behavior, distinguishing between benign and malicious actions more accurately than traditional methods.

4. Automated Threat Hunting

- **Continuous Monitoring:** AI can automate the process of threat hunting, continuously scanning networks, systems, and endpoints for signs of malicious activity. This reduces the manual effort involved in identifying and responding to potential threats.
- **Proactive Threat Detection:** AI tools can proactively search for indicators of compromise (IOCs) across large networks, rather than waiting for threats to be reported or detected by other means.

REAL WORLD EXAMPLE

Some companies using AI in cybersecurity

1.Darktrace

Uses AI for real-time threat detection and automated response. Known for its "Self-Learning AI" that adapts to network behavior.

2.IBM Security (IBM Watson)

Uses AI to analyze threats, identify risks, and help security teams respond faster.

3.CrowdStrike

Uses AI for endpoint protection, detecting malware and ransomware threats quickly. Its platform, Falcon, uses machine learning to stop breaches.

Conclusion:

As cyber threats continue to grow in number and complexity, traditional security methods alone are no longer enough. Artificial Intelligence (AI) is playing a crucial role in strengthening cybersecurity by detecting threats faster, responding in real time, and predicting future attacks. Real-world examples like Darktrace, IBM Watson, CrowdStrike, and Gmail show how AI is already protecting users and organizations worldwide. While AI has its challenges, it is a powerful tool that will shape the future of cybersecurity, making our digital world safer and more secure.

Topology and the Universe's Shape

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For millennia, the universe's vast and mysterious beauty has captivated humanity's imagination. A profound question lies at the heart of our quest to understand the cosmos: What is the shape of the universe? To explore this question, we must delve into the fascinating realm of topology. This is because topology focuses on the qualitative aspects of shapes rather than their precise measurements, allowing for a broader understanding of how different shapes relate to one another in terms of their fundamental characteristics. By examining the universe through the lens of topology, we can better grasp its overall structure and how it may be curved or connected in ways that are not immediately obvious. This approach helps us to uncover the hidden geometry that governs the cosmos, providing insights into the universe's ultimate shape. Topology, often described as "rubber-sheet geometry," is a branch of mathematics that studies properties of shapes that remain unchanged under deformation, such as stretching or twisting, as long as there is no tearing or gluing. For instance, a coffee mug and a donut (torus) are topologically equivalent because one can be deformed into the other without changing their fundamental structure.

In the context of the universe, topology helps us study its global properties and shape. While topology might appear abstract, its applications are remarkably significant and far-reaching. One of the most fascinating applications of topology lies in understanding the shape and structure of the universe. While geometry addresses the local curvature of spacetime, topology analyses the large-scale connectivity of space. Are there boundaries to the universe? Is its extent finite or infinite? Does

space loop back on itself? Topology provides the necessary theoretical structure to address these fundamental questions.

The Universe and Its Geometry

Einstein's theory of General Relativity fundamentally changed our understanding of the cosmos by defining gravity as the curvature of spacetime. According to this theory, the shape of the universe is determined by its total energy density, resulting in three primary geometrical possibilities:

1.Flat Geometry: In a flat universe, Euclidean geometry applies, meaning parallel lines stay parallel and the angles of a triangle add up to exactly 180 degrees. Such a universe could be infinite and unbounded.

2.Spherical Geometry: In spherical geometry, a universe with positive curvature is analogous to a sphere's surface, where parallel lines eventually converge, and sum of angles of a triangle exceed 180 degrees. This model proposes a universe of finite size but without boundaries.

3.Hyperbolic Geometry: In hyperbolic geometry, a universe with negative curvature is saddle-shaped, where parallel lines diverge, and the angles of triangles sum to less than 180 degrees. Similar to a flat universe, a hyperbolic universe could be infinite and unbounded.

While geometry addresses the local curvature, topology extends this understanding of the curvature to the overall structure of the universe.

Topology and the Universe

In cosmology, one of the biggest questions is: What is the universe's shape? Is it infinite or finite? Does it curve, and if so, how? Here, topology provides the mathematical framework for investigating these cosmic enigmas.

1.Global Shape of the Universe: Topologists consider various models for the universe's shape. It might be a simple infinite flat plane, a sphere, or something more complex like a torus (doughnut shape). Understanding the global topology helps cosmologists predict how light and matter behave across cosmic scales.

2.Cosmic Microwave Background (CMB): The CMB radiation is the afterglow of the Big Bang and offers clues about the universe's topology. By analysing patterns in the CMB, scientists infer whether the universe is flat, open, or closed. Some patterns suggest the universe could have a more complex topology, such as a three-dimensional torus.

3.Cosmic Loops and Multi-Connected Spaces: In some topological models, the universe could be finite yet unbounded, like the surface of a sphere or a torus. This implies that traveling in a straight line could eventually bring you back to your starting point. Such concepts challenge our intuitive understanding of space and require deep mathematical reasoning.

Current Evidence and Theories

Data from the Planck satellite and other missions suggest that the universe's geometry is remarkably close to flat. However, this does not mean that there are no unusual topological structures. Although flat, infinite universes with simple topologies remain a prominent model, researchers are always exploring alternative possibilities

One intriguing hypothesis is the "Poincaré Dodecahedral Space," a positive-curved, finite model of the dodecahedron, with its opposite faces connected. This model is consistent with some CMB observations, although further data is needed to validate it.

Conclusion

Topology opens a window to one of the most profound mysteries of existence: the shape and structure of our universe. Through the application of abstract mathematical concepts, scientists can model, predict, and explore the very fabric of space. The combination of topology and cosmology beautifully demonstrates how pure mathematical thought can yield groundbreaking insights into the nature of reality. In this way, topology expands both our mathematical imagination and our understanding of the cosmos. Its application in cosmology is a testament to the power and beauty of mathematical inquiry.

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Edutainment Through Technology: Revolutionizing Today's Education

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Education has always been about creating a bridge between knowledge and the learner. Over the years, traditional methods like chalkboards, textbooks, and lectures have evolved. However, with today's fast-paced world and the rise of digital devices, these old methods often fall short of engaging students. Enter edutainment, a mix of education and entertainment, made even more impactful through technology.

My Perspective on Edutainment

As someone passionate about teaching, I've always believed in making lessons more engaging for students. While I haven't used every tool available, I have seen how technology can transform the way we interact with learners. Instead of the teacher being the sole focus, edutainment encourages a student-centric approach where learners play an active role in their own education.

Interactive tools and creative methods can make lessons easier to understand and more enjoyable. They help shift the focus from passive listening to active participation, where students are involved, curious, and motivated.

What is Edutainment?

Edutainment is simply a way of teaching that blends learning with fun. It's not about replacing lessons with games but enhancing learning by making it more interactive and engaging.

Whether it's using videos, quizzes, or creative design platforms, the goal is to connect with students in ways that capture their attention and imagination.

Why Student-Centric Learning Works

The student-centric approach is built on the idea that students learn better when they're active participants in the classroom. Instead of just absorbing information, they collaborate, explore, and express their ideas. For example, instead of simply explaining a concept, I encourage students to work together on activities or reflect through interactive tools.

Technology plays a key role here. It provides platforms that encourage creativity and two-way communication, ensuring students stay engaged.

IT-Enabled Tools for Interactive Learning

Here are some tools that can make classrooms more engaging and student-focused:

- **Menti.com/Mentimeter** - Mentimeter is a great platform for real-time interaction. Teachers can create polls, word clouds, or quizzes to gauge student understanding during a lesson. It's simple to use and makes students feel heard.
- **Google Forms and Google Classroom**- Google Forms is a quick way to create quizzes, surveys, or feedback forms, while Google Classroom helps manage assignments and share resources. Together, they streamline communication and keep everything organized.
- **EdPuzzle**- EdPuzzle allows teachers to make videos interactive by adding questions or notes. This encourages students to think critically while watching and ensures they stay engaged.
- **Canva/ Prezi** - Canva/ Prezi s a fantastic tool for creativity. Students can design posters, presentations, or infographics, making it ideal for group activities or individual projects. It's simple enough for beginners yet powerful enough to create

professional-looking designs.

- **Kahoot-** This game-based learning platform adds excitement to quizzes. Students love the competitive element, which turns even dry topics into engaging challenges.
- **Padlet-** Padlet acts like an online bulletin board where students can share ideas, links, or images. It's especially helpful for collaborative projects or brainstorming sessions.
- **Google Jamboard-** A digital whiteboard, Jamboard is excellent for discussions and visualizing concepts together. Students can contribute ideas in real-time, whether they're in the classroom or online.
- **Virtual Labs and Simulations** - For management education, simulations replicate real-world scenarios like running a business, managing supply chains, or making financial decisions, mock trading etc. In computer science, virtual labs allow students to practice coding, test algorithms, or explore cybersecurity challenges in a safe, hands-on environment. These tools bridge theory and practice, enhancing learning through real-world applications.
- **Napkin AI-** Napkin AI is an AI-powered platform that converts text into visuals like diagrams, flowcharts, and infographics, simplifying communication and storytelling. It offers customizable visuals with export options in PNG, PDF, and SVG formats, making it ideal for business presentations, education, and reports. Napkin AI enhances understanding by turning complex ideas into clear, engaging visuals.

How Can We Use These Tools Effectively?

Using technology doesn't mean every class needs to be tech-heavy. Instead, it's about balance. Here's how we can integrate these tools in a way that benefits students:

- **Start Small:** Begin with simple tools like Menti.com or Google Forms to introduce interactivity without overwhelming students or teachers.
- **Focus on Activities:** Use tools like Canva, Prezi or EdPuzzle to create student-centric activities. For example, instead of

giving a lecture, ask students to create an infographic or answer questions in a video.

- **Encourage Collaboration:** Platforms like Padlet and Jamboard are perfect for group tasks, where every student can contribute.
- **Seek Feedback:** Regularly ask students what they think about the tools and how they help them learn. This ensures that the methods remain effective and engaging.

Overcoming Challenges

Introducing technology into classrooms can come with its own challenges, such as limited resources or lack of familiarity with the tools. However, these can be addressed with the right approach:

- Choose tools that are free or low-cost, such as Kahoot, Google classroom or Menti.com.
- Provide training for teachers to ensure they feel confident using new platforms.
- Focus on tools that are easy to use, ensuring both teachers and students can adapt quickly.

The Future of Edutainment

Technology in education isn't just a trend—it's becoming a necessity. As educators, it's our responsibility to explore ways to make learning more meaningful. The tools available today can help us explain concepts in ways that are easier for students to understand and remember.

I've realized that when students feel involved in the learning process, they enjoy it more. Technology provides us with the opportunity to create classrooms that are vibrant, creative, and full of life. By embracing these tools, we can transform education into something students look forward to every day.

A Final Thought

Edutainment isn't about replacing teachers or traditional

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methods. It's about using technology to enhance what we already do in classrooms. Whether it's a simple quiz on Menti.com or a creative project on Canva, the goal is to make learning enjoyable and memorable.

To my fellow educators, I say this: don't be afraid to experiment. Start small, focus on your students' needs, and learn as you go. Together, we can create learning environments that truly inspire and empower the next generation.

A Computational Framework for the Perfect Cuboid Problem Using Boolean Filters and Modular Sieves

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The perfect cuboid problem remains one of the most intriguing open questions in number theory. It asks whether a rectangular parallelepiped can exist with all edge lengths, face diagonals, and the space diagonal expressed as integers. While several Euler bricks—configurations with integer edges and face diagonals—are known, no example has ever been found that produces an integer-valued space diagonal. The absence of a confirmed example and the lack of a proof of impossibility leave this mathematical puzzle unresolved.

A perfect cuboid must satisfy four Diophantine conditions simultaneously. For edges a , b , c and diagonals d_{ab} , d_{ac} , d_{bc} and D the system must hold:

$$\begin{aligned}a^2 + b^2 &= d_{ab}^2 \\a^2 + c^2 &= d_{ac}^2 \\b^2 + c^2 &= d_{bc}^2 \\a^2 + b^2 + c^2 &= D^2\end{aligned}$$

Many researchers have attempted to explore this system theoretically and computationally. Earlier surveys trace the historical development of partial solutions and modular restrictions, while more recent computational efforts have extended lower bounds significantly using distributed search techniques and optimized algorithms.

Many researchers have attempted to explore this system theoretically and computationally. Earlier surveys trace the historical development of partial solutions and modular restrictions, while more recent computational efforts have extended lower bounds significantly using distributed search techniques and optimized algorithms.

The present study contributes to this line of work by proposing a computational framework that integrates Boolean logic with modular sieve filtering to accelerate the search for potential perfect cuboids. The goal is not only to explore higher numerical bounds efficiently but also to demonstrate how combining algebraic constraints with logical operations can significantly reduce redundant computation. MATLAB serves as the implementation platform because of its strengths in vectorization, logical evaluation, and parallel processing.

Each of the four Diophantine conditions is encoded as a Boolean expression to quickly verify whether a candidate triple (a,b,c) satisfies the necessary relationships. Boolean expressions such as $a^2 + b^2 = d_{ab}^2$ allow rapid elimination of invalid combinations. However, evaluating these conditions alone across large ranges becomes computationally expensive. To address this limitation, modular arithmetic is used as a preliminary sieve.

Several modular constraints are known for the perfect cuboid problem. Parity restrictions ensure that certain edges must be even, while the space diagonal must be odd. Squares modulo 4, 8, 16, and 19 help eliminate combinations that cannot form perfect squares. Additional divisibility rules apply for primes such as 3, 5, 7, 11, and 19. When combined, these conditions remove over 97% of potential candidates before the Boolean evaluation stage. This drastically lowers computational load and allows searches to extend to larger bounds.

The workflow used consists of generating candidate edge triples, applying modular sieves, computing diagonal values for remaining candidates, and validating them using Boolean checks. MATLAB's vectorization and logical masking make this process efficient, and parfor loops distribute the workload across multiple CPU cores. Integer square verification is carried out using squared-value comparisons instead of floating-point computations to avoid precision errors.

Experiments conducted in MATLAB R202x tested bounds up to $N = 2000$. For $N = 1000$, the modular sieve reduced the candidate set from 10^9 possibilities to 2.5×10^7 , a 97.5% reduction. Similar performance was observed for $N = 2000$. Runtime comparisons also highlight the benefit of the framework: brute-force checking required 42 minutes for $N = 1000$ and nearly 10 hours for $N = 2000$, whereas the modular-plus-Boolean method reduced these times to 9 minutes and 2.3 minutes respectively.

No perfect cuboid was identified within the explored ranges, consistent with all known computational studies. However, known Euler bricks were successfully detected, confirming the correctness of the implementation. The findings reinforce the idea that efficient filtering through modular arithmetic is crucial for extending the computational frontier of this problem. The Boolean-modular approach provides a scalable and flexible foundation upon which additional constraints and theoretical insights may be incorporated.

The study concludes that this hybrid computational model offers significant advantages over brute-force search methods. It opens the door to deeper explorations using GPU acceleration, higher-order modular sieves, and distributed search collaborations. While the perfect cuboid remains elusive, advancements in computational methods such as those demonstrated here strengthen the tools available to mathematicians investigating this centuries-old problem.



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